

- ☒ Master Thesis
- ☐ Bachelor Thesis
- ☐ Research Project

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Algorithmic



Math



Application



Gaze-Adaptive Point Cloud Rendering in VR

Description

Point clouds are promising for VR robot teleoperation because they provide a direct 3D representation of the remote workspace. However, dense point clouds are expensive to render and stream in real time. This thesis investigates a gaze-adaptive rendering strategy for a Varjo VR headset: the region around the user's gaze is rendered with high point density, while peripheral regions are rendered with fewer points. The idea is inspired by foveated rendering for gaze-tracked VR [1,2], but is applied directly to point-cloud density.

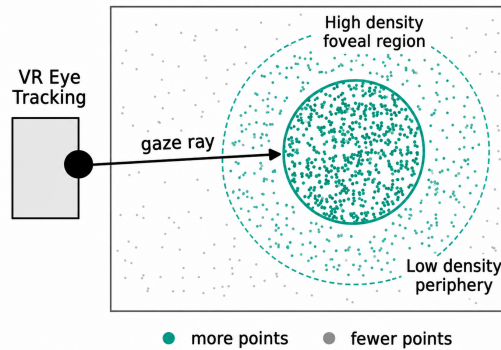


Figure 1: Concept of gaze-adaptive point-cloud rendering. The foveal region is dense; peripheral regions are down-sampled.

Goals and Research Questions

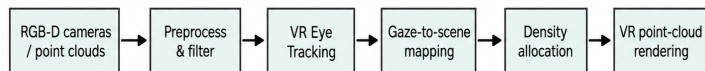
The goal is to build and evaluate a real-time gaze-adaptive point-cloud renderer for VR teleoperation. Main questions:

- Can gaze-adaptive rendering reduce rendered points while preserving task performance and perceived quality?
- How does it compare to full-density, uniformly downsampled, and fixed-center foveated point clouds?
- Which foveal size, density ratio, and smoothing strategy gives the best performance-comfort trade-off?

Tasks

- Eye-tracking integration: Use Varjo gaze data and map the gaze ray into the rendered 3D point-cloud scene.
- Density allocation: Implement high-density foveal, medium-density transition, and low-density peripheral regions.
- Artifact reduction: Smooth gaze updates and density transitions to reduce flickering, popping, and latency effects.
- Baselines: Implement full-density, uniform low-density, fixed-center foveation, and gaze-adaptive foveation.
- User study: Compare the conditions in a VR spatial understanding or teleoperation task.

System pipeline



Study conditions

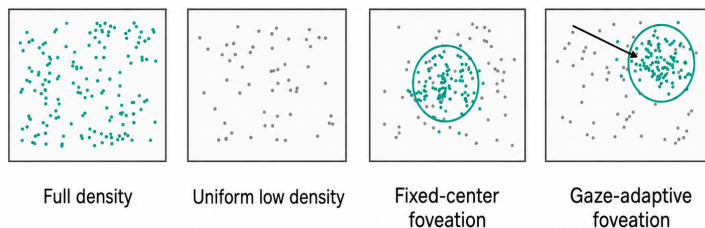


Figure 2: Pipeline and study conditions: full-density, uniform low-density, fixed-center foveation, and gaze-adaptive foveation.

Evaluation

Technical metrics: FPS, latency, GPU load, update rate, and rendered points. User-study metrics: completion time, success rate, accuracy, perceived quality, spatial understanding, NASA-TLX workload [3], SSQ cybersickness [4], and preference ranking.

Expected Outcome

A working gaze-adaptive VR point-cloud renderer and practical design guidelines for using eye tracking to improve XR teleoperation efficiency without reducing spatial awareness or comfort.

[1] Patney et al., ACM TOG, 2016. [2] Guenter et al., ACM TOG, 2012. [3] Hart, HFES, 2006. [4] Kennedy et al., IJAP, 1993.